

Q1: Label **Y** and **Z** as a 1,2-addition product or a 1,4- addition product

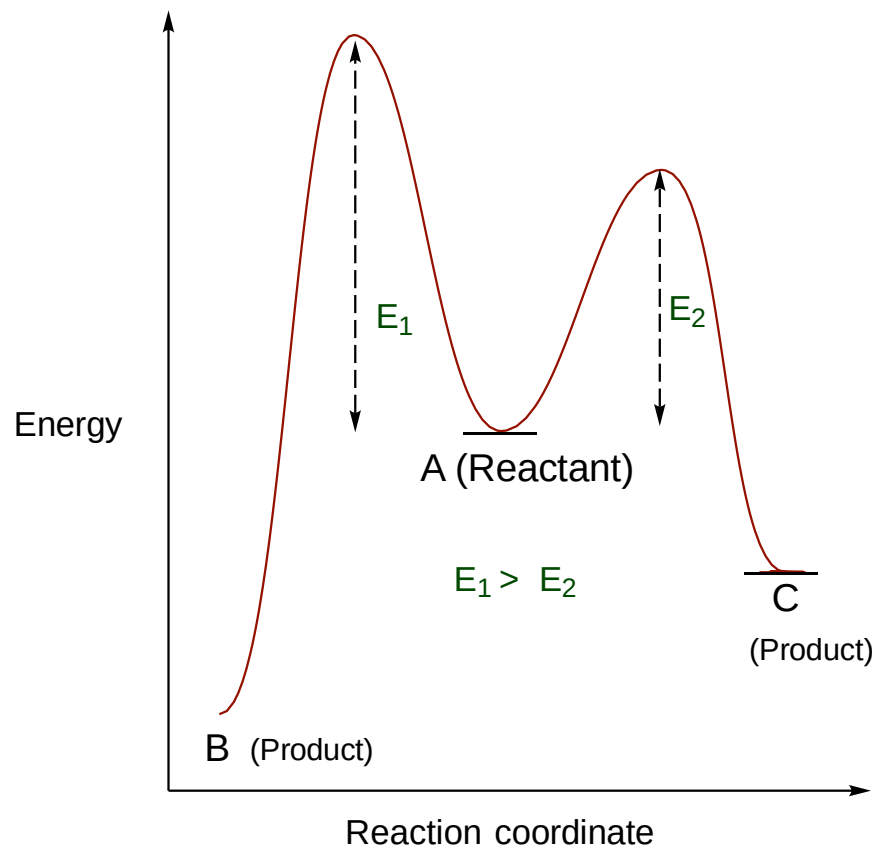
Ans: **Y** is 1,2-addition product and **Z** is 1,4- addition product

Q2: Label **Y** and **Z** as the kinetic or thermodynamic product and explain why.

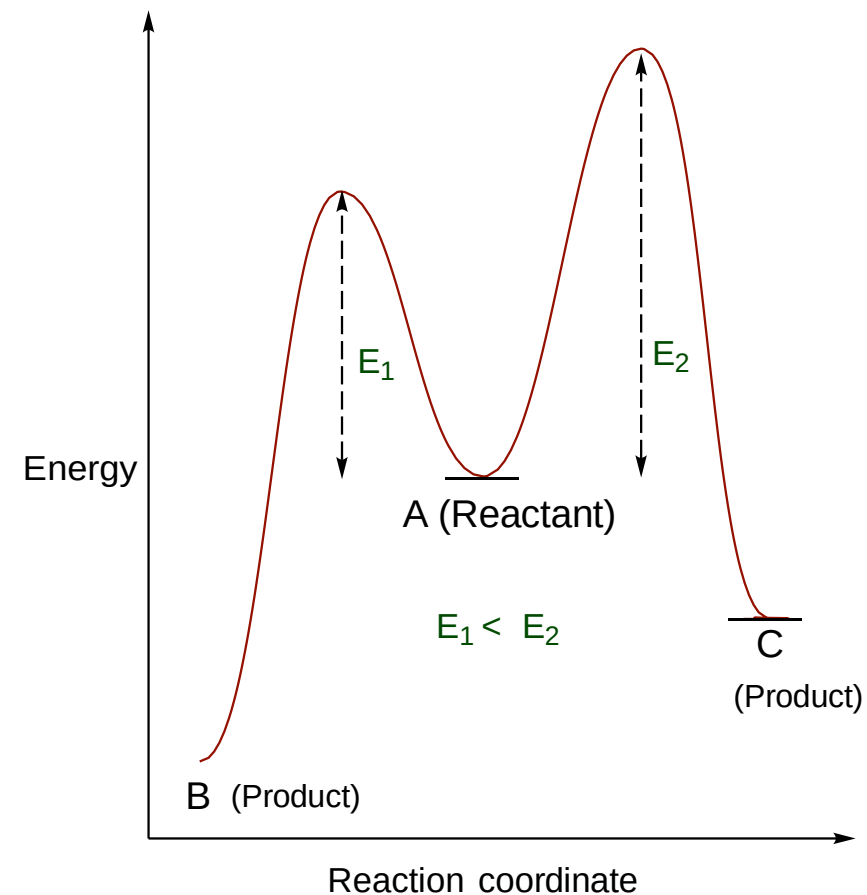
Ans: **Y** is kinetic product and **Z** is Thermodynamic product

Q3: Explain why addition of HCl occurs at the red **C=C**, rather than the blue **C=C**?

Ans: Markovnikov's rule is followed. Addition of H^+ forms the more stable allylic carbocation.



C is the Kinetic product and B is the thermodynamic product

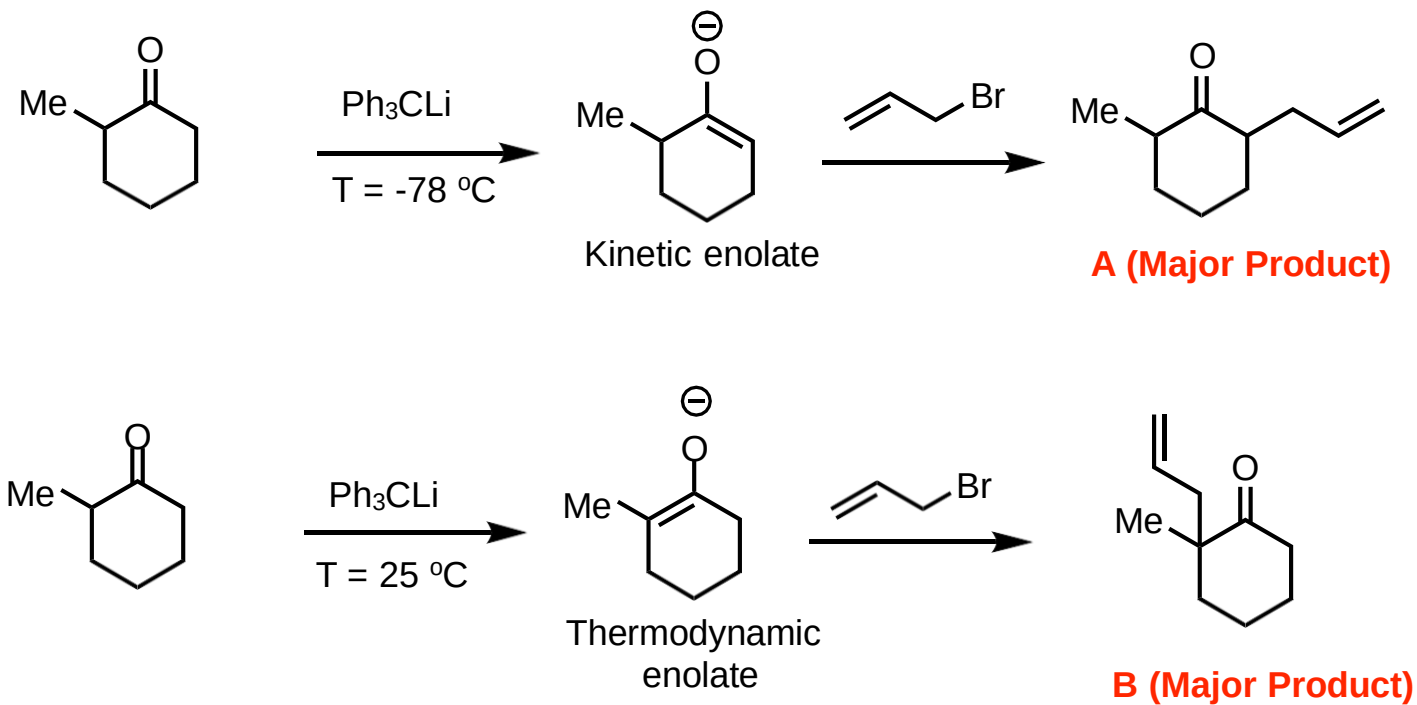


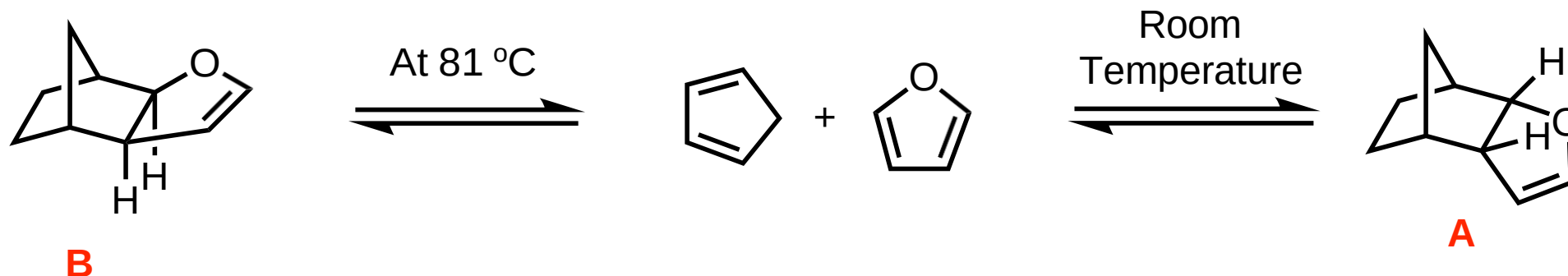
B is both thermodynamic and kinetic product. For these kind of reactions, only trace amount of formation of product C is possible.

Kinetic Product: The the formation of the product requires less activation energy

Thermodynamic Product: The overall energy of the product is lower than kinetic product, i.e, more stable product.

For the deprotonation of an unsymmetrical ketone, the kinetic product is the enolate resulting from removal of the most accessible α -H while the thermodynamic product has the more highly substituted enolate moiety. Use of low temperatures and sterically demanding bases increases the kinetic selectivity. Here, the difference in pK_b between the base and the enolate is so large that the reaction is essentially irreversible, so the equilibration leading to the thermodynamic product is likely a proton exchange occurring during the addition between the kinetic enolate and as-yet-unreacted ketone.





The Diels–Alder reaction of cyclopentadiene with furan can produce two isomeric products. At room temperature, kinetic reaction control prevails and the less stable endo isomer **A** is the main reaction product. At 81 °C and after long reaction times, the chemical equilibrium can assert itself and the thermodynamically more stable exo isomer **B** is formed. The exo product is more stable by virtue of a lower degree of steric congestion, while the endo product is favoured by orbital overlap in the transition state.

